Prep Sheet: §4.4 Undetermined Coefficients & §4.5 Superposition

Active-recall cheatsheet

1 Theory (memorize these; everything else reduces to them)

1.1 The object of study

Constant-coefficient linear 2nd-order ODE:

$$L[y] \;:=\; ay'' + by' + cy \;=\; f(t), \qquad a \neq 0. \tag{NH}$$

Homogeneous associate: $L[y] = 0$. Cited: page 174, eq. (1).

1.2 Structure theorem (why this all works)

Theorem (**General solution decomposition**, p. 182)**.** *If y_p solves (NH) and $\{y_1, y_2\}$ are linearly independent solutions of $L[y] = 0$, then every solution of (NH) has the form*

$$y(t) \;=\; y_p(t) \;+\; c_1 y_1(t) \;+\; c_2 y_2(t).$$

Propositional form. Let $P(y) \equiv L[y] = f$ and $H(y) \equiv L[y] = 0$. Then

$$P(y_p) \wedge H(y_h) \;\vdash\; P(y_p + y_h), \qquad P(y_1) \wedge P(y_2) \;\vdash\; H(y_1 - y_2).$$

The solution set of (NH) is the coset $y_p + \ker(L)$ in the vector space C^2.

1.3 Superposition principle (§4.5, Thm. 3, p. 181)

Theorem. *If $L[y_1] = f_1$ and $L[y_2] = f_2$, then for scalars k_1, k_2:*

$$L[k_1 y_1 + k_2 y_2] \;=\; k_1 f_1 + k_2 f_2.$$

Proof (one line). L is linear in y: substitute and pull scalars out (p. 181).

1.4 Auxiliary equation recap (from §4.2–4.3)

$L[y] = 0 \iff$ trial $y = e^{rt}$ gives the auxiliary equation

$$ar^2 + br + c = 0. \tag{Aux}$$

- Distinct real r_1, r_2: $y_h = c_1 e^{r_1 t} + c_2 e^{r_2 t}$.
- Double real root r: $y_h = c_1 e^{rt} + c_2 t e^{rt}$.
- Complex $\alpha \pm i\beta$: $y_h = c_1 e^{\alpha t} \cos \beta t + c_2 e^{\alpha t} \sin \beta t$.

1.5 Method of Undetermined Coefficients (UC)

Applicability predicate (p. 180): UC works $\iff f(t)$ is a finite sum of atoms of the form

$$C\,t^m\,e^{\alpha t}\cos(\beta t) \quad \text{or} \quad C\,t^m\,e^{\alpha t}\sin(\beta t), \qquad m \in \mathbb{Z}_{\geq 0}.$$

> "The method of undetermined coefficients applies only to nonhomogeneities that are polynomials, exponentials, sines or cosines, or products of these functions." (p. 180).

Anything else (e.g., $\sec t$, $\tan t$, $1/t$, $\ln t$, $t^{-1}e^t$) $\Rightarrow$ UC does not apply; use variation of parameters (§4.6).

Trial-form table (consolidated from p. 178 eqs. (14)–(15), p. 184 eq. (13), p. 185 eq. (14)):

atom $f(t)$	complex index $\rho = \alpha + i\beta$	trial y_p
$Ct^m e^{rt}$	$\rho = r$	$t^s(A_m t^m + \cdots + A_0)e^{rt}$
$Ct^m e^{\alpha t}\cos\beta t$ or $\sin\beta t$	$\rho = \alpha + i\beta$	$t^s\big[P_k(t)e^{\alpha t}\cos\beta t + Q_k(t)e^{\alpha t}\sin\beta t\big]$

where P_k, Q_k are polynomials of degree $k = \max(m_{\cos}, m_{\sin})$ and s is the *multiplicity of ρ as a root of (Aux)*:

$$s = \begin{cases} 0 & \rho \notin \text{roots of (Aux)} \\ 1 & \rho \text{ simple root} \\ 2 & \rho \text{ double root (only possible if } \beta = 0) \end{cases}$$

Why the t^s? If ρ is a root, $e^{\rho t} \in \ker(L)$, so plugging $Ae^{\rho t}$ into L returns 0 and A is lost (p. 176–177, eqs. (11)–(13)). Each factor of t reduces the "resonance defect" by one order (p. 178, Cases 1–3).

1.6 Master algorithm

Algorithm (Undetermined Coefficients + Superposition)**.** *Input: $a, b, c, f(t)$.*

1. *Solve $ar^2 + br + c = 0$; record root set R with multiplicities.*
2. *Decompose $f(t) = \sum_k f_k(t)$ where each f_k is a single UC atom.*
3. *For each f_k:*
   a. *Read off $\rho_k = \alpha_k + i\beta_k$ and polynomial degree m_k.*
   b. *Set $s_k := \mathrm{mult}_R(\rho_k) \in \{0, 1, 2\}$.*
   c. *Build $y_{p,k}$ per trial-form table.*
4. *$y_p := \sum_k y_{p,k}$. Substitute into $L[y_p] = f$, match like-term bases, solve linear system for the A_j, B_j.*
5. *General solution: $y = y_p + c_1y_1 + c_2y_2$. Apply ICs if given (§4.5, Thm. 4, p. 182).*

Debugging invariants.

- *Invariant A:* The trial y_p must include *every* power $t^m, t^{m-1}, \ldots, t^0$ even if f only contains t^m (p. 175).
- *Invariant B:* Sine forces cosine companion and vice versa (p. 175, eq. (6)).
- *Invariant C:* Grouping identical trial forms is legal and preferred; "streamlined" matching (p. 184 Ex. 4).

2 §4.4 Prep Problems

2.1 §4.4 #5 — Classify: does UC apply?

Model problem M5. Decide if UC applies to

$$y''(x) + 2y'(x) + y(x) = \ln x.$$

Solution. $\ln x$ is not a polynomial, exponential, sine, cosine, or a product thereof. By the applicability predicate (p. 180), **UC does not apply**. Use variation of parameters.

Your problem (#5).

$$y''(\theta) + 3y'(\theta) - y(\theta) = \sec\theta.$$

[Hint: check the predicate.]

2.2 §4.4 #7 — Classify: does UC apply?

Model problem M7. Decide if UC applies to

$$5u'(x) - 2u(x) = 7x^{50}e^{-2x}\sin(3x).$$

Solution. $f(x) = \underbrace{7x^{50}}_{\text{polynomial}} \cdot \underbrace{e^{-2x}}_{\text{exponential}} \cdot \underbrace{\sin(3x)}_{\text{sine}}$. Product of three UC-atoms $\Rightarrow$ itself a UC-atom with $m = 50$, $\alpha = -2$, $\beta = 3$. **UC applies.** (UC extends to any-order constant-coefficient linear ODE; the §4.4 first-order case is identical in form.)

Your problem (#7).

$$8z'(x) - 2z(x) = 3x^{100}e^{4x}\cos(25x).$$

2.3 §4.4 #11 — Find a particular solution

Model problem M11. $y'' - y = 3^t$.

Solution. Rewrite $3^t = e^{t\ln 3}$, so $\rho = \ln 3$, $m = 0$. Aux: $r^2 - 1 = 0 \Rightarrow r = \pm 1$. Since $\ln 3 \approx 1.0986 \notin \{\pm 1\}$, $s = 0$.

Trial $y_p = Ae^{t\ln 3} = A \cdot 3^t$. Then $y_p'' = A(\ln 3)^2 \cdot 3^t$, so

$$y_p'' - y_p = A\left[(\ln 3)^2 - 1\right]3^t = 3^t \implies A = \frac{1}{(\ln 3)^2 - 1}.$$

$$\boxed{y_p(t) = \frac{3^t}{(\ln 3)^2 - 1}.}$$

Your problem (#11). $y''(x) + y(x) = 2^x$. [Hint: $2^x = e^{x\ln 2}$; aux roots $\pm i$; $\ln 2$ is not a root.]

2.4 §4.4 #17 — Find a particular solution (resonance case)

Model problem M17. $y'' + 9y = 6\cos(3t)$.

Solution. Aux: $r^2 + 9 = 0 \Rightarrow r = \pm 3i$. Atom gives $\rho = 0 + 3i = 3i$, which *is* a simple root, so $s = 1$, $m = 0$. Trial:

$$y_p = t\big(A\cos 3t + B\sin 3t\big).$$

Compute:

$$\begin{aligned} y_p' &= A\cos 3t + B\sin 3t + t(-3A\sin 3t + 3B\cos 3t) \\ y_p'' &= -6A\sin 3t + 6B\cos 3t - 9t(A\cos 3t + B\sin 3t) \\ y_p'' + 9y_p &= -6A\sin 3t + 6B\cos 3t = 6\cos 3t. \end{aligned}$$

Match: $-6A = 0$, $6B = 6 \Rightarrow A = 0, B = 1$. $\boxed{y_p(t) = t\sin 3t.}$

Your problem (#17). $y'' + 4y = 8\sin 2t$. [Hint: same pattern; $\rho = 2i$ is a root; expect $y_p = \alpha t\cos 2t$ for some α.]

2.5 §4.4 #23 — Polynomial RHS, zero is a root

Model problem M23. $y''(t) - 3y'(t) = t^3$.

Solution. Aux: $r^2 - 3r = 0 \Rightarrow r = 0, 3$. RHS $= t^3 e^{0\cdot t}$, so $\rho = 0$, $m = 3$. Since 0 is a *simple* root of (Aux), $s = 1$.

$$y_p = t(A_3t^3 + A_2t^2 + A_1t + A_0) = A_3t^4 + A_2t^3 + A_1t^2 + A_0t.$$

$$\begin{aligned} y_p' &= 4A_3t^3 + 3A_2t^2 + 2A_1t + A_0, \\ y_p'' &= 12A_3t^2 + 6A_2t + 2A_1, \\ y_p'' - 3y_p' &= -12A_3t^3 + (12A_3 - 9A_2)t^2 + (6A_2 - 6A_1)t + (2A_1 - 3A_0) = t^3. \end{aligned}$$

Solve:

$$A_3 = -\tfrac{1}{12}, \quad A_2 = -\tfrac{1}{9}, \quad A_1 = -\tfrac{1}{9}, \quad A_0 = -\tfrac{2}{27}.$$

$$\boxed{y_p(t) = -\tfrac{t^4}{12} - \tfrac{t^3}{9} - \tfrac{t^2}{9} - \tfrac{2t}{27}.}$$

Your problem (#23). $y''(\theta) - 7y'(\theta) = \theta^2$.

2.6 §4.4 #27 — Form only (no coefficient evaluation)

Model problem M27. Write the form of a particular solution to

$$y'' + 4y = 5t^2\cos 2t.$$

Solution. Aux: $r^2 + 4 = 0 \Rightarrow r = \pm 2i$. Atom has $\rho = 2i$, which *is* a root, so $s = 1$. $m = 2$. By (15) on p. 178 with $k = \max(m_{\cos}, m_{\sin}) = 2$:

$$\boxed{y_p(t) = t(A_2t^2 + A_1t + A_0)\cos 2t + t(B_2t^2 + B_1t + B_0)\sin 2t.}$$

Your problem (#27). $y'' + 9y = 4t^3\sin 3t$. [Hint: $\rho = 3i$ is a root of $r^2 + 9 = 0$; $s = 1$, $k = 3$.]

3 §4.5 Prep Problems

3.1 §4.5 #1 — Using superposition

Model problem M1. Given

- $u_1 = \frac{t}{2}$ solves $y'' + 2y = t$,
- $u_2 = \frac{e^{-t}}{3}$ solves $y'' + 2y = e^{-t}$.

Find particular solutions to:

(a) $y'' + 2y = 6t$: RHS $= 6 \cdot t \Rightarrow y_p = 6u_1 = 3t$.
(b) $y'' + 2y = 2t + 9e^{-t}$: RHS $= 2 \cdot t + 9 \cdot e^{-t} \Rightarrow y_p = 2u_1 + 9u_2 = t + 3e^{-t}$.
(c) $y'' + 2y = -4t + 6e^{-t}$: $y_p = -4u_1 + 6u_2 = -2t + 2e^{-t}$.

Rule of thumb. Read the scalar k_i directly off each RHS atom; apply Thm. 3 (p. 181).

Your problem (#1). $y_1 = \cos t$ solves $y'' - y' + y = \sin t$; $y_2 = e^{2t}/3$ solves $y'' - y' + y = e^{2t}$. Solve for RHS = (a) $5\sin t$, (b) $\sin t - 3e^{2t}$, (c) $4\sin t + 18e^{2t}$.

3.2 §4.5 #7 — Build general solution from given y_p

Model problem M7. Given $y_p(t) = -t^2$ is a particular solution of

$$y'' - 4y = 4t^2 - 2,$$

find the general solution.

Solution. Verify: $(-t^2)'' - 4(-t^2) = -2 + 4t^2 = 4t^2 - 2$. ✓ Aux: $r^2 - 4 = 0 \Rightarrow r = \pm 2$. So $y_h = c_1e^{2t} + c_2e^{-2t}$.

$$\boxed{y(t) = -t^2 + c_1e^{2t} + c_2e^{-2t}.}$$

Your problem (#7). $y'' = 2y + 2\tan^3 x$, $y_p(x) = \tan x$. Find general solution.
[Hint: rewrite as $y'' - 2y = 2\tan^3 x$; aux $r^2 - 2 = 0$; this is the homogeneous eq., not the nonhomog. RHS.]

3.3 §4.5 #9 — Classify with superposition

Model problem M9. Can UC + superposition find a particular solution of

$$y'' - y' + 5y = 3e^{2t} + t\sin(4t) - 7?$$

Solution. Decompose RHS:

$$f_1 = 3e^{2t}, \quad f_2 = t\sin 4t, \quad f_3 = -7.$$

Each is a UC atom (exponential; polynomial×sine; constant). By Thm. 3, $y_p = y_{p,1} + y_{p,2} + y_{p,3}$.
Yes, UC+superposition applies.

Your problem (#9). $3y'' + 2y' + 8y = t^2 + 4t - t^2e^t\sin t$. [Hint: three atoms t^2, $4t$, $-t^2e^t\sin t$; all UC-valid.]

3.4 §4.5 #21 — Full general solution with resonance

Model problem M21. $y'' + 2y' + 5y = e^{-t}\cos 2t$.

Solution.

1. Aux: $r^2 + 2r + 5 = 0 \Rightarrow r = -1 \pm 2i$. So $y_h = c_1 e^{-t}\cos 2t + c_2 e^{-t}\sin 2t$.
2. RHS atom: $\rho = -1 + 2i$, which *is* a root. $s = 1$, $m = 0$.
3. Trial: $y_p = Ate^{-t}\cos 2t + Bte^{-t}\sin 2t$.
4. **Shortcut computation.** Let $f(t) = te^{-t}$. Then

$$f' = (1-t)e^{-t}, \quad f'' = (t-2)e^{-t}, \quad f'' + 2f' + f = 0.$$

Apply $L = D^2 + 2D + 5$ to $f(t)\cos 2t$ and $f(t)\sin 2t$:

$$\begin{aligned} L[f\cos 2t] &= \underbrace{(f'' + 2f' + f)}_{=0}\cos 2t + (-4f' - 4f)\sin 2t \\ &\qquad + \text{ correction } = -4e^{-t}\sin 2t, \\ L[f\sin 2t] &= \qquad\qquad 4e^{-t}\cos 2t. \end{aligned}$$

(Derivation: expand $(fg)''$ with $g = \cos 2t$ or $\sin 2t$; terms with $f'' + 2f' + f$ vanish because $(D+1)^2[te^{-t}] = 0$; surviving cross-terms give $\pm 4e^{-t} \cdot$ (swap of sin/cos); see analogous treatment in Ex. 3, p. 183.)

5. Set $L[y_p] = A \cdot L[f\cos 2t] + B \cdot L[f\sin 2t] = -4Ae^{-t}\sin 2t + 4Be^{-t}\cos 2t \stackrel{!}{=} e^{-t}\cos 2t$.

$$4B = 1, \ -4A = 0 \implies A = 0, \ B = \tfrac{1}{4}.$$

6. $y_p = \frac{t}{4}e^{-t}\sin 2t$.

$$\boxed{y(t) = c_1 e^{-t}\cos 2t + c_2 e^{-t}\sin 2t + \tfrac{t}{4}e^{-t}\sin 2t.}$$

Your problem (#21). $y''(\theta) + 2y'(\theta) + 2y(\theta) = e^{-\theta}\cos\theta$.
[Hint: aux roots $-1 \pm i$; $\rho = -1 + i$ is a root; $s = 1$. Expected answer shape: $y = c_1 e^{-\theta}\cos\theta + c_2 e^{-\theta}\sin\theta + \frac{\theta}{2}e^{-\theta}\sin\theta$.]

4 Quick-reference summary card

Decision tree.

1. Is RHS a sum of $t^m e^{\alpha t}\{\cos\beta t, \sin\beta t\}$ atoms? **No** → use §4.6 variation of parameters. **Yes** → continue.
2. Compute roots of (Aux).
3. For each atom: read off $\rho = \alpha + i\beta$, m; set $s = \mathrm{mult}_R(\rho)$.
4. Build trial per table. Substitute, match, solve linear system.
5. Add y_h. Apply ICs if any.

Two traps.

- Forgetting t^s when ρ is a root $\Rightarrow$ inconsistent linear system.
- Dropping sine (or cosine) companion $\Rightarrow$ inconsistent linear system.